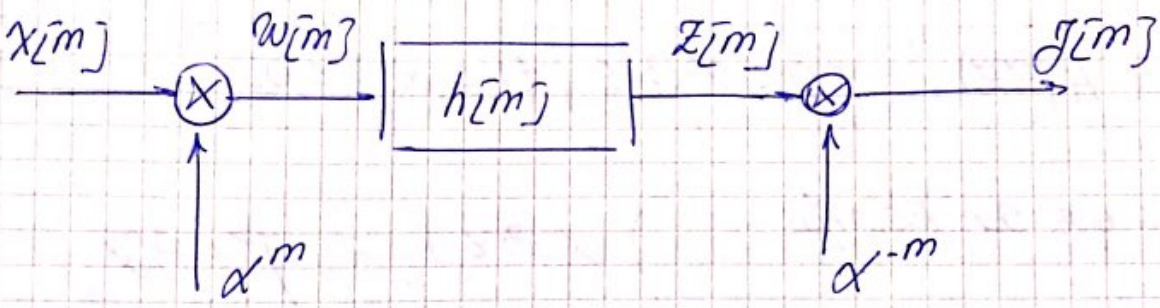
11 AGOSTO DE 2014

1)



a) ROC: $0 \leq \beta \leq |z| \leq \gamma < \infty$

PARA QUE $h[m]$ SEA ESTABLE, LA ROC DEBE CONTENER A LA CIRCUNFERENCIA UNIDAD

$\Rightarrow \beta < 1 \wedge \gamma > 1$

b) $w[m] = \alpha^m x[m] \Rightarrow W(z) = X(z/\alpha)$

$z[m] = w[m] * h[m] \Rightarrow Z(z) = W(z)\,H(z)$

$g[m] = \alpha^{-m} z[m] \Rightarrow Y(z) = Z(\alpha z)$

$\Rightarrow Y(z) = W(\alpha z)\,H(\alpha z)$

$Y(z) = X(z)\,H(\alpha z)$

$H(\alpha z) \rightarrow h'[m] = \alpha^{-m} h[m]$

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c) PARA $H(z)$: $0 \leq \beta \leq |z| \leq \gamma < \infty$

PARA $H(\alpha z)$: $0 \leq \alpha\beta \leq |z| \leq \alpha\gamma < \infty$.

PARA QUE SEA ESTABLE

- $\alpha\beta < 1 \Rightarrow \beta < 1/\alpha$
- $\alpha\gamma > 1 \Rightarrow \gamma > 1/\alpha$

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$x[m]$ LARGO N ; $X[k]$ DFT DE $x[m]$

$$w[m] = \frac{1}{2} + \frac{1}{2}\cos\left(\frac{2\pi}{N}(m - N/2)\right)$$

SEA: $$g[m] = x[m]\,w[m] = \frac{1}{2}x[m] + \frac{1}{2}x[m]\cos\left(\frac{2\pi}{N}(m-N/2)\right)$$

AHORA BIEN: $$e^{j\frac{2\pi}{N}(m-N/2)} = e^{j\frac{2\pi}{N}m}e^{-j\pi} = -e^{j\frac{2\pi}{N}m}$$

$$= -\left\{\cos\left(\frac{2\pi}{N}m\right) + j\,\text{SEN}\left(\frac{2\pi}{N}m\right)\right\}$$

EN REALIDAD:

$$\cos\left(\frac{2\pi}{N}(m-\tfrac{N}{2})\right) = \frac{e^{j\frac{2\pi}{N}(m-\frac{N}{2})} + e^{-j\frac{2\pi}{N}(m-\frac{N}{2})}}{2}$$

$$\cos\left(\frac{2\pi}{N}(m-\tfrac{N}{2})\right) = \frac{e^{j\frac{2\pi}{N}m}e^{-j\pi} + e^{-j\frac{2\pi}{N}m}e^{j\pi}}{2}$$

$$\cos\left(\frac{2\pi}{N}(m-\tfrac{N}{2})\right) = \frac{-e^{j\frac{2\pi}{N}m} - e^{-j\frac{2\pi}{N}m}}{2}$$

ENTONCES:

$$g[m] = \frac{1}{2}x[m] + \frac{1}{2}x[m]\left\{\frac{-e^{j\frac{2\pi}{N}m} - e^{-j\frac{2\pi}{N}m}}{2}\right\}$$

$$g[m] = \frac{1}{2}x[m] - \frac{1}{4}e^{j\frac{2\pi}{N}m}x[m] - \frac{1}{4}e^{-j\frac{2\pi}{N}m}x[m]$$

$$g[m] = \frac{1}{2}x[m] - \frac{1}{4}W_N^{-1m}x[m] - \frac{1}{4}W_N^{1m}x[m]$$

$$\therefore\ Y[k] = \frac{1}{2}X[k] - \frac{1}{4}X[(k-1)_N] - \frac{1}{4}X[(k+1)_N]$$

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2) $E_c = \int_{-\infty}^{+\infty} |x_c(t)|^2 dt < \infty \Rightarrow \frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(j\omega)|^2 d\omega$

$= \frac{1}{2\pi} \int_{-W}^{W} |X(j\omega)|^2 d\omega < \infty$

MUESTREANDO: $x[n] = x_c(nT)$

$X(e^{j\Omega}) = \frac{1}{T} X_c\left(j\frac{\Omega}{T}\right)$

SEA: $\Omega = \omega T \Rightarrow d\Omega = d\omega T \Rightarrow d\omega = \frac{d\Omega}{T}$

$\Rightarrow E_c = \frac{1}{2\pi} \frac{1}{T} \int_{-\pi}^{\pi} \left|X_c\left(j\frac{\Omega}{T}\right)\right|^2 d\Omega$

2 CASOS:

a) SIN ALIASING: $\omega_s = \frac{2\pi}{T} \geq 2W \Rightarrow \boxed{T \leq \pi/W}$

$\Rightarrow E_D = E_c/T$

b) CON ALIASING:

$E_D = \frac{1}{2\pi} \frac{1}{T} \int_{-\pi}^{\pi} \left|X_c\left(j\frac{\Omega}{T}\right)\right|^2 d\Omega + \frac{1}{2\pi} \frac{1}{T} \int_{-\pi}^{\pi+2} \left|X_c\left(j\frac{\Omega}{T}\right)\right|^2 d\Omega$

EFECTO DE BORDE

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